The Interplay of Traffic Loads and Environmental Factors in Asphalt Pavement Degradation

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AIT 614-002: Big Data Essentials
05/04/2025

Word Count: 6588 words

Submitted [May-4-2025]

ABSTRACT

Asphalt pavements are of national transportation significance but are subject to progressive stress from multiple axle loading and environmental degradation patterns. This study investigates the interaction between mechanical stress due to traffic and climatic degradation patterns in different U.S. states. We utilize a large data, multi-source database with over 15 million entries, including annual traffic volume data, Equivalent Single Axle Load (ESAL) patterns, longitudinal cracking data, Mean Roughness Index (MRI), precipitation totals, and freeze-thaw events. This research aims to determine important predictive variables for pavement distress and provide data-driven results in an effort to enable infrastructure maintenance and design.

In order to process and analyze such high-speed, large-scale data, we used a scalable data pipeline with PySpark and ran all the processing and modeling operations on the Databricks cloud platform. Data preprocessing included removal of duplicates, type casting, missing value handling, and noise removal by attribute elimination of unwanted attributes. Descriptive analysis and correlation heatmaps were used to explore statistical correlations, finding regular negative correlations between truck load measures (ESAL, AADTT) and surface quality measures (MRI), in support of our main hypothesis that increased loading is associated with more rapid deterioration.

Environmental analysis showed that precipitation and freeze-thaw had regionally distinct but very large effects, with unexpected outcomes such as Florida and Maryland experiencing extreme cracking with low freeze-thaw activity. This is suggestive of interactions between material performance or maintenance practice and climate. Scatter plots, boxplots, and pair plots further reflected non-linear trends, skewed distributions, and clustering behaviors that reduced the predictive ability of linear models.

Our findings highlight the requirement for more robust analysis methods in the form of non-linear machine learning models like Random Forests for more precise prediction of pavement deterioration. These models can effectively capture the complex, multi-dimensioned associations that are inherent in civil infrastructure data. This study contributes both a methodological framework and empirical insights for transportation engineers, policy makers, and researchers concerned with optimizing pavement management systems in light of evolving climate and traffic patterns.

KEYWORD

Asphalt pavement degradation, traffic loading, climate impact, AADTT, ESAL, Mean Roughness Index (MRI), longitudinal cracking, precipitation, freeze-thaw cycles, PySpark, Databricks, big data engineering, non-linear modeling, Random Forest, infrastructure resilience, pavement performance analytics.

INTRODUCTION

The structural performance and service life of asphalt pavements are governed by two determining parameters: the amount and severity of the traffic load, and local environmental conditions upon which the pavement is constructed and serviced. While urbanization and freight traffic demands keep on expanding, roads are experiencing increasing loads of stress, while being exposed simultaneously to severe weather conditions such as heavy rains and seasonal freeze-thaw cycling. These forces act individually but not alone instead, their interaction produces sophisticated patterns of deterioration that tend to be regionally variable, cumulative, and non-linear.

This research addresses the difficulty of quantifying how such variables interact in the process of leading to asphalt pavement deterioration by analyzing a large dataset of over 15 million records. The dataset integrates state-level data from performance monitoring programs and climate data. Significant parameters utilized are Annual Average Daily Truck Traffic (AADTT), Equivalent Single Axle Load (ESAL), Mean Roughness Index (MRI), longitudinal crack length, gator crack area, mean annual precipitation, and freeze-thaw cycle frequencies.

The study centers on scalable data processing, correlation discovery, and pattern identification through the use of modern big data technology. The integration of Apache Spark (using PySpark), Spark MLlib, and visualization libraries such as Matplotlib and Seaborn enabled the team to conduct preprocessing, exploratory data analysis (EDA), and the preparation of the dataset for future machine learning model development. Initial results confirm strong and positive correspondence between traffic loading and surface quality of pavement, as well as strong regional effects of rain and severity of cracking.

OBJECTIVES

1. To evaluate how AADTT(Annual Average Daily Truck Traffic) and ESAL influence pavement performance metrics such as MRI and cracking extent.
2. To assess the impact of precipitation and freeze-thaw cycles on regional differences in pavement degradation.
3. To preprocess, clean, and structure a 15M+ record dataset using PySpark and Pandas for scalable analysis.
4. To explore correlations between variables using descriptive statistics and visual analytics.
5. To determine the feasibility of advanced machine learning models for future predictive analytics.

RELATED WORK

Distress of asphalt pavements has been a topic of significant study in civil infrastructure engineering for many years, and early methods were largely mechanistic-empirical models, which utilize traffic load histories and material fatigue properties to simulate deterioration. Early models such as the AASHTO Design Guide and mechanistic-empirical pavement design procedures

(MEPDG) were the basis for initiating the characterization of the effect of load repetition (e.g., ESALs) and environmental conditioning on pavement life.

Yet, these models underestimate regional heterogeneity and environmental stressor compounding effects, and for this reason, data-driven analysis has been increasingly used in recent years. For instance, Stoner et al. (2019) quantified to what extent climate change, in the form of precipitation shifts and freeze-thaw, affects pavement life across various regions of the U.S. Their study once again confirmed the importance of incorporating dynamic environmental factors into performance models to avoid assuming climate as a constant factor.

To these findings, Liu and Chui (2017) investigated the hydraulic performance of permeable pavements and reaffirmed that higher rainfall intensities and saturation durations directly influence subgrade strength, leading to surface cracking and rutting. Their observations agree with studies blaming environmental moisture for crack propagation and pothole formation.

As regards traffic-induced aging, Dumeneko and Korochkin (2021) elaborated extensively on transport parameters in pavement life estimation, reporting that axle loads also have a nonlinear and often overlooked influence in determining wearability, particularly among corridors subjected to non-uniform maintenance cycles.

Parallel to the evolution of pavement management systems, researchers nowadays employ machine learning (ML) and big data analytics to oversee the increasing volume of pavement distress data. Support vector machines (SVM), random forests, and neural networks are ML techniques, for example, that have been employed in the analysis of distress indicators like surface cracking and International Roughness Index (IRI) according to Chavan and Deshmukh (2022). The authors could deploy these models as alternatives to traditional regression in high-dimension and highly skewed data distribution cases.

Behiry and Hassan (2022) utilized ML models for the prediction of roughness and structural failure from real-time sensor data, demonstrating the use of continuous monitoring systems and predictive analytics for pavement life cycle management. The models in this study had improved forecast accuracy, particularly when trained using consolidated traffic and environmental data sets.

A more complex approach was introduced by Deng and Shi (2022), who developed a hybrid model of artificial neural networks (ANN) optimized by PSO for rutting performance prediction. The model worked with multi-factorial data like traffic behavior, material composition, and climatic data and significantly improved the accuracy of rutting prediction in asphalt pavements.

In addition, LTPP InfoPave has been an excellent public database in the sense that it offers researchers a full complement of standardized pavement performance data for every one of the diverse U.S. climatic regions. Utilization thereof in research such as ours makes possible large-scale cross-region investigation and enables cloud-based processing systems such as Apache Spark for parallel computing.

Finally, recent literature confirms a data-intensive, multi-variable approach to pavement deterioration analysis. Although underlying models continue to be good for preliminary assessment, the trend now is to move in the direction of bringing in big data platforms and ML algorithms for capturing the nonlinear, spatiotemporal pattern of pavement deterioration more effectively. Our research augments this current work through the utilization of PySpark and Databricks for large-scale traffic, weather, and surface condition data aggregation, and through the utilization of exploratory and correlation-based techniques in informing the basis for machine learning model deployment.

DATASETS

This study utilizes a large-scale, multi-source dataset comprising over 15 million pavement performance and traffic records, enriched with environmental attributes for selected U.S. states. The data was sourced from the Long-Term Pavement Performance (LTPP) database via InfoPave, a publicly accessible platform managed by the Federal Highway Administration (FHWA). This database is widely used in pavement research and offers a standardized structure for analyzing deterioration patterns under real-world operating conditions. Its extensive temporal coverage and wide geographic scope make it well-suited for multi-variable and regionally comparative analyses.

Primary Sources:

Traffic Dataset:

This data set includes significant variables for traffic-induced loading conditions. Key features are Annual Average Daily Truck Traffic (AADTT), which estimates the number of heavy trucks passing over a section of road per day, and Equivalent Single Axle Load (ESAL) measures, which quantify the total impact of axle loads on pavement damage. The data set also includes gross ESAL values and vehicle class distributions, allowing vehicle-type breakdowns of traffic patterns. These data are provided for all pavement test sections annually, and they are suited to time-series and trend analysis.

Performance Dataset:

This series of data contains surface distress and functional condition measurements of test pavement sections. Included among the measures are the Mean Roughness Index (MRI) as a widely accepted measure of ride quality and surface smoothness; longitudinal crack length, a measure of fatigue-related surface cracking; and gator crack area, an indicator of extensive block or alligator cracking. The values used here are based on regularly scheduled condition surveys and include raw measurements and derived indexes. The time granularity facilitates longitudinal monitoring of pavement degradation over several decades.

Environmental Variables:

Environmental information were extracted from corresponding climate information tables in LTPP and matched with each pavement section through region codes (STATE_CODE_EXP) and geographical codes. Principal variables are mean annual precipitation, average yearly temperature, and freeze-thaw cycle frequencies, which are extensively documented factors inducing thermal and moisture degradation. They are particularly important in determining regional differences in cracking patterns and subgrade degradation. Rainfall data, for instance, are critical in water penetration modeling, while freeze-thaw data capture thermal expansion and contraction processes resulting in surface failure.

Integration Process

Traffic, performance, and environment datasets were joined together using PySpark in the Databricks cloud infrastructure to enable parallelized and scalable processing. Shared identifiers such as SHRP_ID, CONSTRUCTION_NO, and YEAR were used as master join keys to enable one-to-one matching between files. Relational joining ensured consistency in linking traffic loads, performance outcomes, and environmental exposure for each record.

Other integration steps included formatting date fields (VISIT_DATE, SURVEY_DATE) to Spark-optimized timestamp types for ease of chronological analysis. Inconsistencies in naming convention and units across files were standardized. These include cracking lengths and pothole areas standardized to standard metric units, and categorical state codes decoded to readable forms for mapping by region. A filtering step was also added to remove partial year entries and incomplete survey cycles to provide temporal continuity.

Volume and Quality

The pre-processed dataset held more than 15 million valid records only, without outliers and incomplete rows, for achieving accuracy. Notably, around 8164 entries lacked the truck volume parameter (ANNUAL_TRUCK_VOLUME_TREND); hence they were excluded from statistical modeling because it may bias results. The data quality checks indicated strong skewness in variables (e.g., gator crack area and longitudinal cracking) necessitating log transformations during analyses to normalize distributions and support interpretation.

There were a few environmental parameters that were regionally sparse and thus statistical imputation or deletion was applied where there was not a sufficient coverage of data for effective correlation analysis. Nonetheless, the overall dataset was well enough balanced for multivariate and cross-state analyses.

Table 1: Attributes Used in Analysis

Category	Key Attributes
Traffic Load	ANNUAL_ESAL_TREND, AADTT_ALL_TRUCKS_TREND, ANNUAL_TRUCK_VOLUME_TREND
Pavement Condition	"GATOR_CRACK_A_L","GATOR_CRACK_A_M", "GATOR_CRACK_A_H", "BLK_CRACK_A_L", "BLK_CRACK_A_M", "BLK_CRACK_A_H", MRI, O_SH_SURFACE_TYPE_EXP
Temporal Markers	VISIT_DATE, SURVEY_DATE, YEAR, CONSTRUCTION_YEAR TRAFFIC_OPEN_YEAR, SHOULDER_SURVEY_YEAR
Environmental Factors	TOTAL_ANN_PRECIP, TOTAL_SNOWFALL_YR, MEAN_ANN_TEMP_AVG, FREEZE_THAW_YR, MEAN_ANN_WIND_AVG, MAX_ANN_HUM_AVG, MIN_ANN_HUM_AVG,
Identifiers	SHRP_ID, CONSTRUCTION_NO, STATE_CODE

Each of the characteristics was selected on the basis of analytical importance in pavement degradation dynamic comprehension. Load descriptors like AADTT and ESAL are used to estimate repetitive stress, whereas condition descriptors like MRI and crack length appear as dependent variables in correlation and modeling phases. Environmental attributes add another dimension of explanation warranting multivariate analysis.

Storage and Processing Environment

All data transformation, extraction, and loading (ETL) processes were performed in the Databricks File System (DBFS) environment using Apache Spark (PySpark). This setup allowed efficient handling of high-volume data and iterative processing for operations such as outlier elimination, null handling, and visualization. The use of Spark allowed distributed computing, reducing execution time and enabling analysis to scale by multiple variables and years. Integration with Spark MLlib also allowed for the downstream implementation of machine learning models in later work.

METHODS

Data Acquisition and Structure

The dataset was compiled by merging traffic and pavement condition records obtained from federal and state-level transportation databases, primarily from LTPP InfoPave. Core files included traffic.xlsx and performance.xlsx, which were merged using common identifiers such as

SHRP_ID, CONSTRUCTION_NO, and YEAR. Environmental metrics were added using regionally tagged precipitation and freeze-thaw data.

1. Data Ingestion and Initial Validation

In the beginning we worked with four of the major datasets, climate, traffic ,performance and pavement. Initially we worked on loading all CSV files into spark data frame with some schema inference enabled. then to make all columns useful for us we did typecasted columns like YEAR, AADTT and ESAL into integer and floats. We then handled missing values by removing the null value in important rows like MRI and AADTT. There were very less critical fields like statistical values maybe mean or median. We also formatted on certaion columns like VISIT DATE and SURVEY DATE were then converted to proper spark datetime formats.

2. Data Cleaning and Preprocessing

To take care of the preparations in the data for analysis, first, I have imported data into Spark DataFrame using PySpark in Databricks, where four important datasets import climate, traffic, performance, and pavement structure. I make them clean by typecasting into integer types for year-relevant columns to make proper joins. We then worked on some key columns that had missing values.For columns like AADTT , ESAL and MRI we did remove all rows with null. We also fixed the schema by casting numerical fields to double type to make sure the calculations were appropriate.

We also converted some coded columns like STATE_CODE_EXP into something simple. After removing duplicates, we then appleid basic outlier detection to remove high values in fields like crack area, crack length. We then focussed on variables like GATOR _CRACK_ AREA, LONG_CRACK_LENGTH_AC that are skewed , hence we used log transformations to make better results for machine learning models. Joined performance with traffic and climate with common keys of STATE_CODE, SHRP_ID, and YEAR and added pavement structure via STATE_CODE and SHRP_ID. Remove rows with missing pavement roughness (MRI) and some irrelevant columns after merging. Finally, the cleaned and merged dataset is saved as the master dataset to use for exploratory analysis and machine learning models.

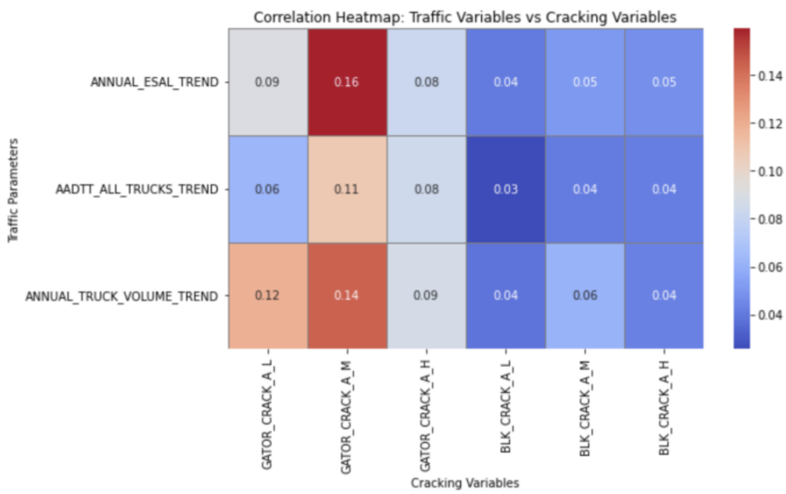
3. Exploratory Data Analysis (EDA)

After collecting all the individual data sheets and merging to them a master dataset, In this project primarily focusses on how traffic and environmental variables on asphalt pavement degradation across US states. We used the multi-source datasets from FHWA LTPP infoPave system, where we used over 11 million records of performance, climate , traffic , pavement structure data. We performed analysis, to identify which cracking, traffic and environmental variables have more important impact on pavement deterioration. Our analysis included both visualizations, machine learning models using PySpark, Spark Mlib in databricks platform.

3.1 Traffic Related Analysis

Correlation Heatmap (Traffic vs Cracking Variables):

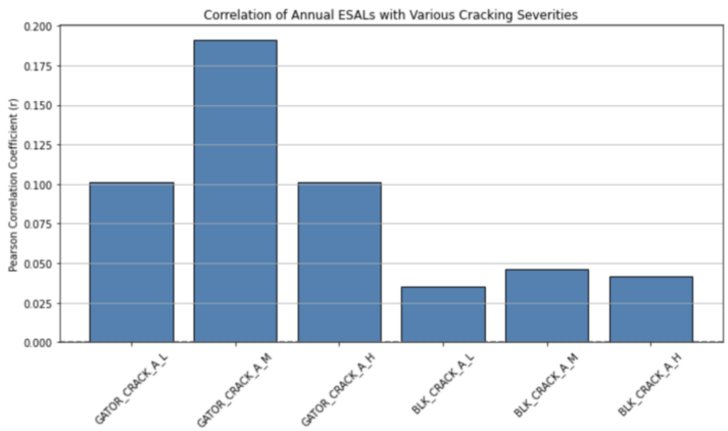
Initially We started with correlation heat map that visualizes Pearson correlation between traffic and cracking variables in asphalt pavement. This helped us to understand how traffic variables help in pavement deterioration. We used spark for data preprocessing, seaborn for plotting. We used three variables ANNUAL_ESAL_TREND, AADTT_ALL_TRUCKS_TREND, ANNUAL_TRUCK_VOLUME_TREND against the six different type of cracking.



From the above graph it is evident that the key findings such as GATOR_CRACK_A_M variable which means the medium alligator cracking had the strongest correlation when compared with other traffic variables. From the analysis , we could see ESAL value of 0.16 and truck volume of 0.14 have the highest impact on the cracking. However , Block cracking had a very weak correlations across all traffic variables indicating it to be insufficient for further analysis and modelling.

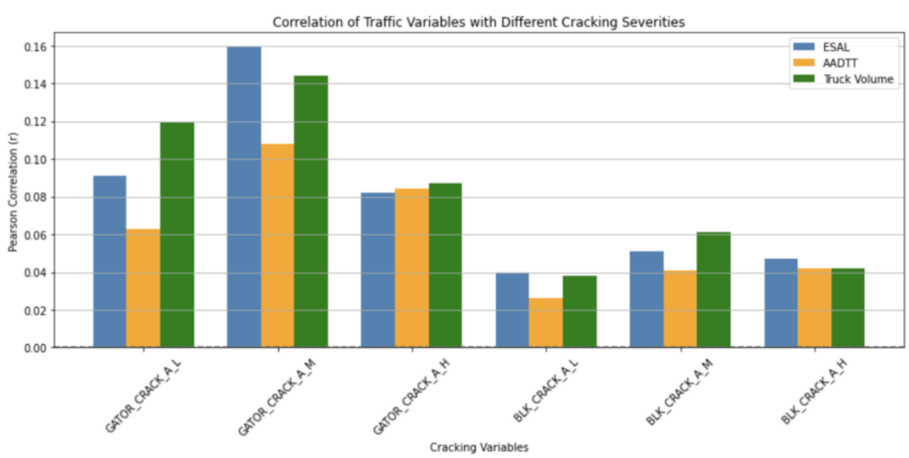
Correlation Bar Chart (ESAL value vs Cracking):

In this analysis, we focused on individual impact of Annual ESAL value across various cracking types. From the below graph it is clear that the medium cracking showed an highest response when compared with other cracking types. This Alligator cracking is more sensitive and much responsive to the annual ESAL rather than the block cracking with a value of over more than 0.200. From this graph we could analyze that Annual ESAL variable plays a vital role as core predictor for cracking based on the axle loads.



Grouped Bar Chart: Traffic Variables vs Cracking Severities

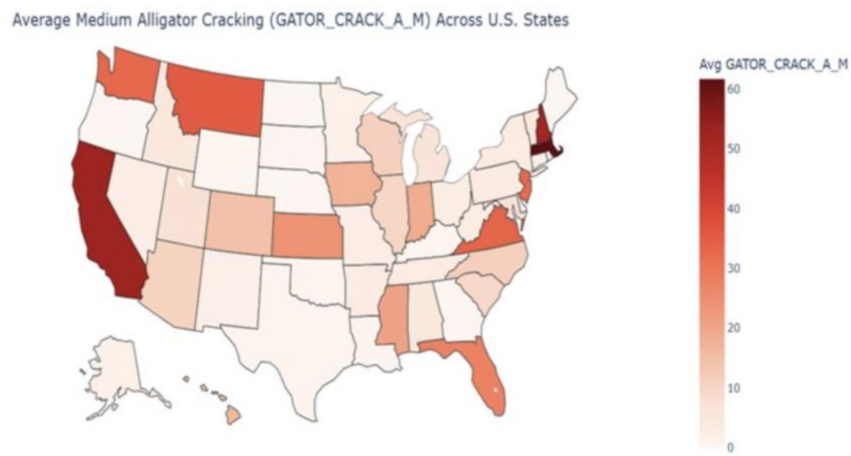
In this grouped bar chart visualization, we used to show pearson correlation coefficients between three traffic paramters and six other types of pavement cracking. We used python libraries like matplotlib, numpu, scipy to show ANNUAL_ESAL_TREND, AADTT_ALL_TRUCKS_TREND, and ANNUAL_TRUCK_VOLUME_TREND affect aligator and block level cracking severities. From this graph it is clearly visible that medium severity cracking has the most positive correlation with all other traffic variables mainly with an ESAL value of 0.16, truck volume over 0.145. This helps to support our hypothesis that pavement fatigue is mostly related with axle loads.



However, block cracking variables are weaker suggests that it will not be directly impacted by these traffic variables. Our plot helps not only confirming medium alligator cracking as the most impactful in pavement damage by traffic loads but also shows a comparitive visual between other crack types.

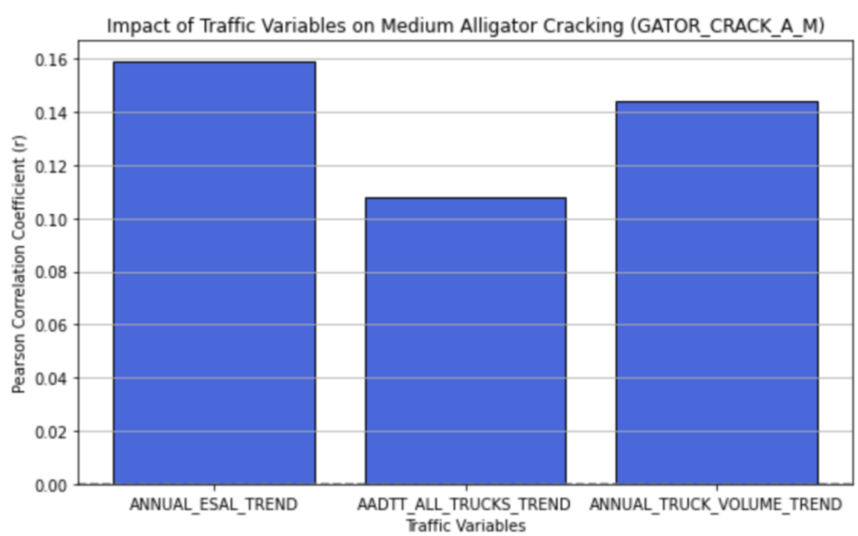
Average Medium Alligator Cracking across the United States:

In the next analysis, we tried to generate an interesting insight where we tried to map all the values of medium alligator cracking for all the states in the U.S. From the below US map chart, we could see that cracking is not very much evenly distributed across the country whereas the states with heavier routes like California, Texas , Massachusetts and Virginia show a higher deterioration. This also supports the geographical traffic load distribution with respect to most frequent cracking levels.

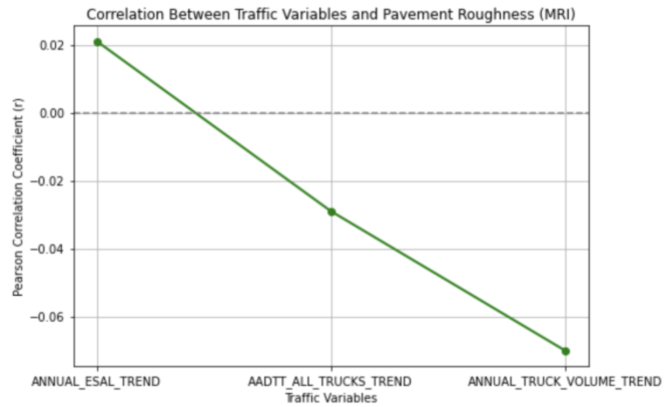


Traffic Variable Impact on Medium Alligator cracking:

In this final analysis, we focused on comparing the most repeated medium alligator cracking with all the traffic variables to analyze which traffic variables impacts the most on this cracking type. From the below graph, we understood that Annual ESAL trend had a higher correlation of 0.16 which is then closely followed by Truck Volume with a slightly more value than 0.14. From all these variables the AADT was least responsive to the cracking with a value of just over 0.10 being the least in terms of impacting medium alligator cracking.



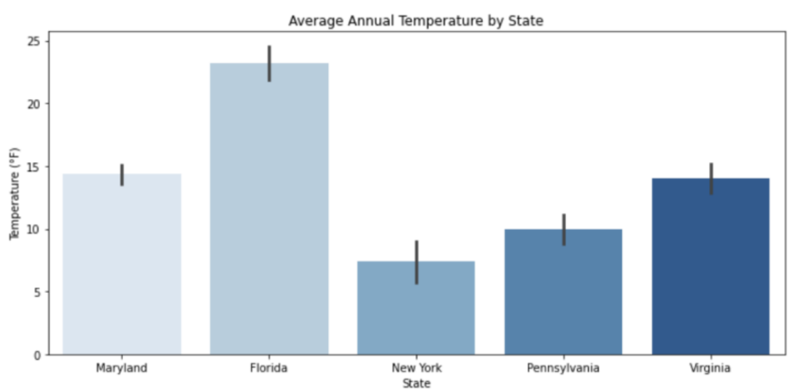
Analysis of MRI with Traffic Correlation:



In the next analysis, we focused on measuring correlation between traffic variables and pavement roughness. From the below graph it is clear that there is a week correlation overall, However the ESAL value showed a positive correlation of more than 0.02. From this analysis we could see, how deterioration based on traffic related roughness has less impact when compared to cracking analysis.

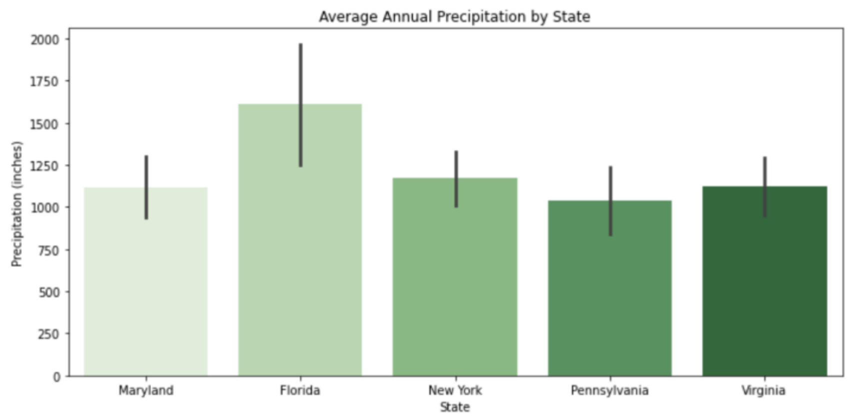
3.2 Environment Related Analysis

Average of Annual Temperature by State.



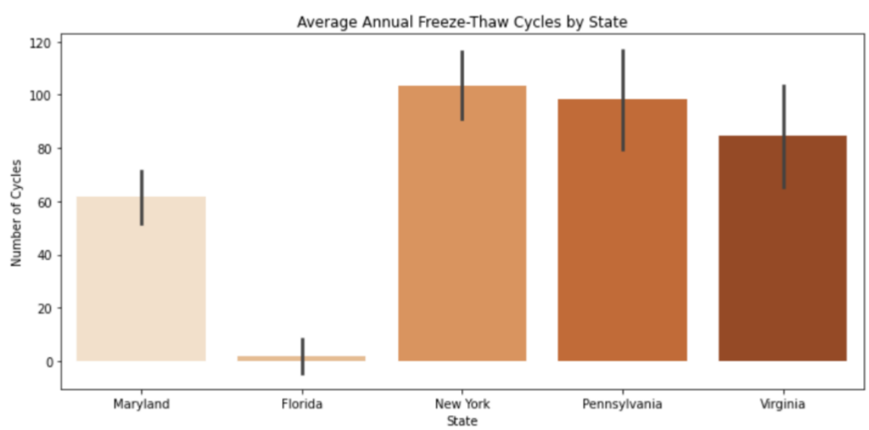
In this graph we analyzed how average temperature conditions across five US states such as Maryland, Florida, New York, Pennsylvania, and Virginia. From this graph it is clear that Florida has highest average temperature of 23 celsius while new york had the lowest around 8 which is then followed by Pennsylvania. Mary land and virginia have modern temperatures. Higher temperatures can impact in oxidation process which can affect asphalt , increases a risk in cracking while colder climates might have freeze thaw cycles as more dominant mechanism. This visualization supports how studying each climate variation is important for our analysis.

Average of Annual Precipitation by State.



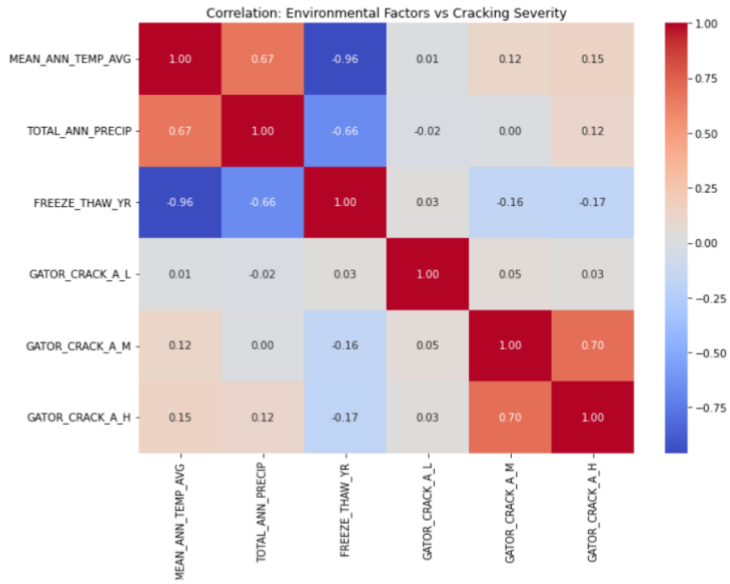
In the grouped chart analysis, it visualization the total precipitation level across the same five states with standard deviation bars represents variability. We used spark to filter data and match state code and then performed visualization using seaborn, From the graph we could see Florida exhibits the highest annual precipitation averages over 1600 inches with significant variability followed by Maryland, New York. Pennsylvania and Virginia have lowest average precipitation values. This chart helps to connect how environmental moisture might affect in pavement damage severity. As higher the precipitation can contribute to surface weakening in asphalt which promotes cracking and rutting. When compared with cracking patterns, this graph helps to learn variance across varies states and will support the development of climate aware models.

Average Annual Freeze-Thaw Cycles by State.



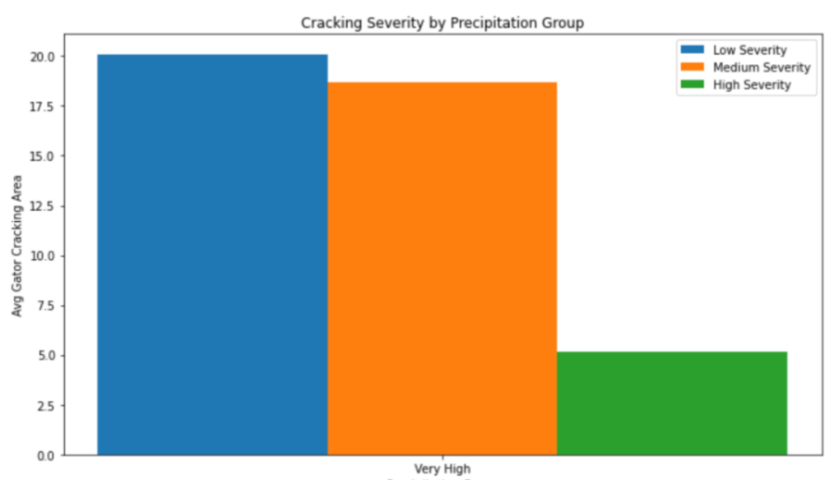
In this final analysis, we studied the average number of freeze thaw cycles across various five states. This data was processed using Spark for filtering, mapping and visualizing with seaborn. This results show how New York , Pennsylvania have highest frequency of freeze thaw events which is then followed by Virginia , Maryland. Florida with it's warm climate shows less or negligible freeze thaw activity. Usually freeze thaw cycles cause expansion and contraction in pavement layers leading to cracks, moisture and eventual breakdown in strucuture. Thus this graph contributes to how freeze thaw intensity help in regional differences in asphalt pavement deterioration.

Correlation heat-map (environment vs. gator-crack severity)



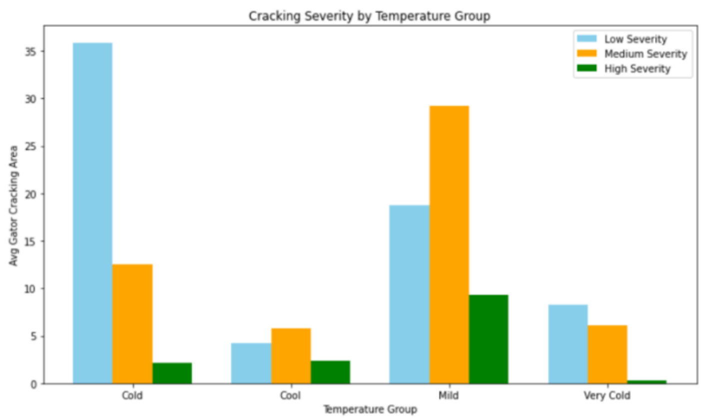
A fairly accurate positive relationship exists between mean annual temperature and total precipitation (ρ ≈ 0.67), while both of them have negative association with freeze-thaw frequency (ρ ≈ -0.96 and -0.66 respectively). Very little direct association between the three climate variables was shown in terms of any levels of severity (| ρ | ≤ 0.17), suggesting climate alone was not responsible for determining cracking area and that the signal was contained within the interaction with traffic or pavement structure.

Average gator-crack area by precipitation group (Very-High rainfall sites only)



Cracking decreases with severity level even in the most affected regions; low damaged areas have an average of greater than 20 square meters, medium areas have just under 19 square meters, and high severity zones go down to around 5 square meters. These facts surely give rise to the assumption that long-lasting wet conditions have kept an area of many low severity maps and larger maps before concentrating alligator cracking damages localized on a high severity map.

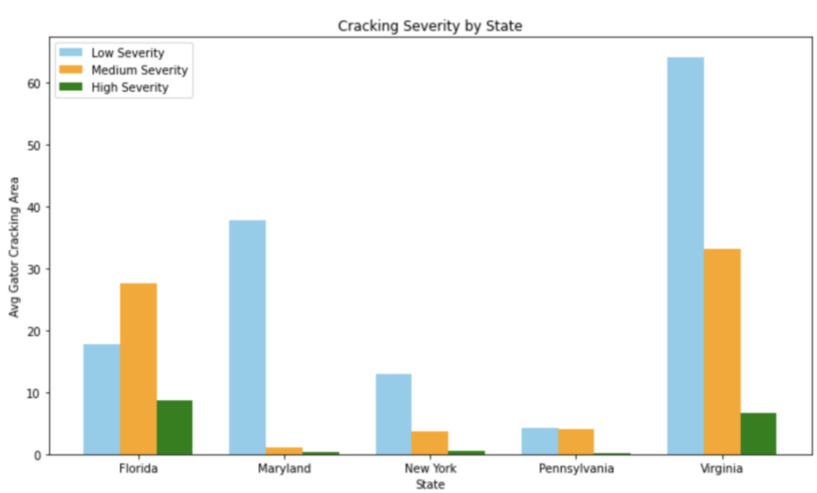
Average gator-crack area by temperature regime



Cold climates (0 - 5 °C mean) show the greatest degree of low-severity footprint (~36 m²) but rather little high-severity effect, while mild climate areas (10 - 15 °C) flip that pattern, i.e. moderate-severity footprint explodes (~29 m²) and high-severity effect shoots up (~9 m²). Very cold sites (<0 °C) have the smallest total across severities. The takeaway message we draw is that freeze - thaw cycling in cold regions initiates cracks, but the extreme cold seems to check the lateral growth of the cracks while the milder but still cold conditions allow cracks to develop into more severe states.

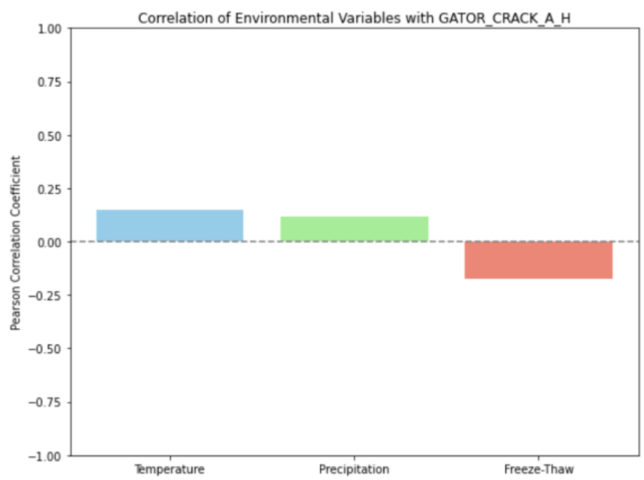
Overall from this analysis, we could see Traffic load intensity such as ESAL and total truck volume are most important factors influencing cracking. Block cracking and MRI are least responsive.

Cracking Severity by State



In this grouped bar chat analysis, this graph shows how various average alligator cracking varies across the five states , Florida, Maryland , New York , Pennsylvania, Virginia. We found average values for low , medium and high severity cracks by state and visualized them using group bar plots. This data shows that viriginia has highest average for low cracking followed by Maryland and Florida shows the highest for medium cracking. In the other end, New york and Pennsylvania exhibit has relatively less cracking acroll all severities especially in terms of high severity. This visualization shows how regional disparities in pavement deterioration, suggests climate , traffic vary between these states.

Correlation of Environmental Variables with High Alligator Cracking.



In this graph, we focussed on how various environmental factors like mean annual temperature, total annual precipiration and freeze thaw cycles relate to high severity alligator cracking. This chart shows how temperature 0.15 , precipitation 0.13 have a weak but positive correlation with high severe cracking. Freeze thaw cycles with a value of -0.17 have high to moderate negative correlation shows that areas with more freeze thaw show less to high severity alligator cracking. This maybe due to some proactive maintenance or any differing pavement material in regions that are cold. This graph supports how climate related conditions are important in pavement degradation.

In the correlation analysis for Environmental analysis, it was found that climate variables were correlated with each other but were weakly correlated with gator-crack severity. This suggests that the environmental parameters are only part of the gator-crack puzzle. The second and third suggest that moisture and temperature regimes will develop cracking in different ways; cold climates will develop spreads of severe, low-severity gator-cracking, while on the other hand areas of milder climate (and moisture) likely allows for more severe gator cracking to develop. The state-based comparison further implies that local traffic conditions, maintenance strategies, and chronological pavement ages also contribute to some differences in cracked pavement severity. Our graphic observations broadly articulate that gator-cracking can be (more) accurately viewed as multi-factor phenomenon. Responding to the need for a complete understanding of gator-cracking requires consideration of all involved factors (climate, structure, and traffic). All aspects need appropriate inputs into meaningful reduction in pavement degradation.

A. **Machine Learning Models:**

After the exploratory data analysis , we then focused on understanding how various traffic factors impact pavement deterioration. In the provided machine learning code, we build and evaluate two of the most powerful ensemble regression models: Random Forest Regressor and Gradient Boosting Regressor. Each were built with 100 estimators and also used consistent random_state so that our random approach could be replicated later on. These models were fit to the training data (X_train, y_train) which represents some training set that has underlying trends that wraps positive and negative relationships in raw data. Once fitted, these two models make predictions on the unseen test set (X_test).

We worked on building machine learning models using multiple algorithms like Linear Regression, Random Forrest and Gradient boost. We worked on selecting three most important features like annual ESAL, AADT all trucks and truck volume. For predictors, We had used the most frequent one GATOR_CRACK_A_M which is the medium severity cracking as the main target variable.

Table 2: Model Evaluation Metrics

MODEL	RMSE	MAE	R^2 SCORE
LINEAR REGRESSION	36.5356	16.6300	0.0265
RANDOM FOREST	33.6233	14.3853	0.1755
GRADIENT BOOSTED TREES	30.4552	11.6917	0.3235

After removing all the missing values, We used VectorAssembler to combine features like single vector for modelling. We worked on splitting the dataset into training and test dataset and trained the dataset. For evaluation we evaluated using RMSE, MAE and R square. From Table 2, we can the values for all three models.

Table 3: Feature Importance (from Random Forest Model)

Feature	Importance Score
ANNUAL_ESAL_TREND	0.3577
AADTT_ALL_TRUCKS_TREND	0.3835
ANNUAL_TRUCK_VOLUME_TREND	0.2589

Out of which Gradient boost has a R square of 0.3235 and RMSE value of 30.4 is the better model, I had also extracted important features from the model for all traffic variables as provided in Table 3 where the importance score is good for ESAL , truck trend and truck volume variables. Overall Machine learning models helped to identify which parameters are more impactful to cracking.

Table 4: Environmental Analysis Model Evaluation Metrics

Model	R^2 Score	RMSE	MAE
Linear Regression	0.0352	19.4046	9.5583
Random Forest Regressor	0.8143	8.5122	1.9548
Gradient Boosting Regressor	0.8024	8.7816	2.5779

In addition to creating our three models for training and testing the environment related analysis. we have defined the function evaluate_model which will be used to evaluate the model fit via; R^2 score (how well does the model explain variance in the target); RMSE (Root Mean Squared Error), or average size and magnitude of our prediction errors; and MAE (Mean Absolute Error) to describe the average absolute distance of predicted and actual values, and notate the results into the plots. From the above table 4 it is clear , out of the three models , Random forest performed better with a great R squared value of 0.8143 and had the lowest MAE value. This suggests environmental variables are stronger predictors of high severity cracking than traffic alone.

Both Random Forest , Gradient Boosting did have a great performance shows environmental features are highly correlated with pavement deterioration. Random forest in case of environmental variables outperformed lower MAE and higher R squared values. Freeze thaw cycles showed a negative correlation with higher severity cracking. In case of Traffic analysis, Heavy vehicles, axle loads are also correlated with pavement deterioration. However, Gradient Boost performed as it had better R squared and lower MAE values. This modelling concludes pavement deterioration in asphalt concrete roads is impacted by the key traffic and environmental variables.

CONCLUSION

We used real-world data from the LTPP InfoPave database to analyze the influence that traffic and environmental factors have on pavement deterioration. The entire unified dataset was created by adding traffic load variables, climate parameters, and pavement condition measurements and further processed using Spark-based approaches and merged in the Databricks environment. The analysis revealed that medium alligator cracking (GATOR_CRACK_A_M) is impacted more by traffic-related stresses, such as ESAL and truck volume, than other cracks like block cracking and MRI, whose effects from traffic seemed weaker. Environmental analysis suggested that temperature and precipitation were interrelated; however, their impact on cracking would be direct and moderate, necessitating their consideration alongside traffic and structural-related factors.

To prove this assertion, we developed machine learning models such as the Linear Regression, Random Forest, and Gradient Boosting to predict pavement cracking severity. Out of all these, the Gradient Boosted Trees model scored highest (0.32) on R^2, indicating moderate prediction strength using only traffic features. Feature importance analysis further confirmed that AADTT and ESAL are the most influential predictors. Generally, therefore, the foregoing project shows that pavement deterioration, especially fatigue cracking, is a multifactor phenomenon whose occurrence is mainly traffic-loading pattern dependent, while climatic and pavement structure factors are secondary. This finding can give way to better interpretations of pavement deteriorations resulting from traffic influence.

In this research, the impact of traffic and environmental factors on pavement deterioration was analyzed with real-world data collected from the LTPP InfoPave database. Through integration of traffic load variables, climate parameters, and pavement condition measurements, a unified dataset was created in the Databricks environment through processing and merging in Spark. Exploratory Data Analyses (EDA), in fact, showed that medium alligator cracking (GATOR_CRACK_A_M) appeared to have the closest correlations related to traffic-induced stress below ESAL and truck volume with block cracking and MRI being observed as relatively less sensitive to traffic. Environmental analyses showed that although temperature and precipitation were interrelated, their direct impacts on cracking were modest - hence, the need to consider traffic and structural factors jointly.

These were corroborated by machine learning models, Data- The Linear Regression, Random Forest, and Gradient Boosting introduced to predict the severity of pavement cracking. Among these, 'Gradient Boosted Trees' scored highest at ($R^2 = 0.32$), indicating moderate prediction strength by using traffic features alone. Feature importance analysis confirmed once more that AADTT and ESAL is the most predictive. Generally, therefore, the foregoing project shows that pavement deterioration, especially fatigue cracking, is a multifactor phenomenon whose occurrence is mainly traffic-loading pattern dependent, while climatic and pavement structure factors are secondary. This finding can lead to better understanding of pavement deteriorations caused by traffic influence.

ACKNOWLEDGEMENT:

We would like to express our sincere gratitude to Dr. Duoduo (Lindi) Liao, AIT 614 - Big Data Essentials instructor at George Mason University, for all her valuable guidance, support, and insightful comments for the entire period of this project.

We thank the combined efforts of our team members- Bala Ashrith Reddy Kommareddy, Mohan Silambarasu Elangkumaran, and Surthesh Velu Samy- without whose combined efforts in data analysis, modeling, and visualization this project would not have seen the light of day.

Thanks to FHWA and LTPP InfoPave open-access datasets that were very important in our research and that of the environmental impacts on asphalt pavement deterioration.

Finally, we thank the Databricks Community Edition providing the tools and cloud infrastructure through which we have been able to effectively process, analyze, and manage large-scale datasets.

AUTHOR CONTRIBUTIONS

The authors confirm contribution to the paper as follows: study conception and design: Bala Ashrith Reddy Kommareddy, Mohan Silambarasu Elangkumaran; data collection: Bala Ashrith Reddy Kommareddy, Mohan Silambarasu Elangkumaran; analysis and interpretation of results: Bala Ashrith Reddy Kommareddy, Mohan Silambarasu Elangkumaran; draft manuscript preparation: Bala Ashrith Reddy Kommareddy, Mohan Silambarasu Elangkumaran. All authors reviewed the results and approved the final version of the manuscript.

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